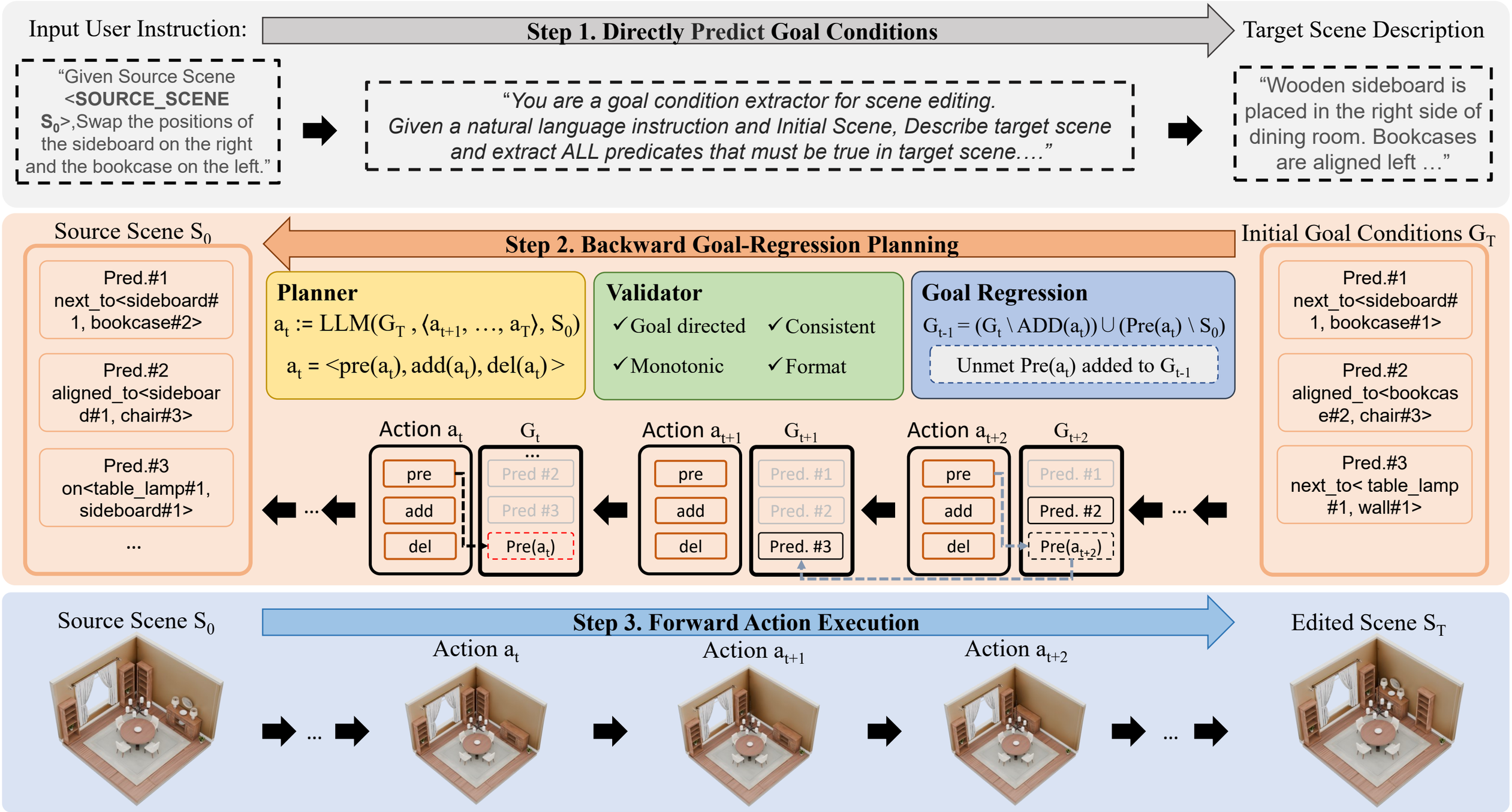
Input User Instruction:
Step 1. Directly Predict Goal Conditions
Target Scene Description
"Given Source Scene <SOURCE_SCENE S0>,Swap the positions of the sideboard on the right and the bookcase on the left."
"You are a goal condition extractor for scene editing. Given a natural language instruction and Initial Scene, Describe target scene and extract ALL predicates that must be true in target scene…."
"Wooden sideboard is placed in the right side of dining room. Bookcases are aligned left …"
Source Scene S0
Step 2. Backward Goal-Regression Planning
Initial Goal Conditions GT
Pred.#1 next_to<sideboard#1, bookcase#2>
Pred.#2 aligned_to<sideboard#1, chair#3>
Pred.#3 on<table_lamp#1, sideboard#1>
...
Planner
at := LLM(GT , ⟨at+1, …, aT⟩, S0)
at = <pre(at), add(at), del(at)>
Validator
✓Goal directed ✓Consistent
✓Monotonic ✓Format
Goal Regression
Gt-1 = (Gt \ ADD(at)) ∪ (Pre(at) \ S0)
Unmet Pre(at) added to Gt-1
Action at
Gt
pre
add
del
Pred #2
Pred #3
Pre(at)
Action at+1
Gt+1
Pred. #1
Pred. #2
Pred. #3
Action at+2
Gt+2
Pre(at+2)
Pred.#1 next_to<sideboard#1, bookcase#1>
Pred.#2 aligned_to<bookcase#2, chair#3>
Pred.#3 next_to< table_lamp#1, wall#1>
Source Scene S0
Step 3. Forward Action Execution
Edited Scene ST
Action at
Action at+1
Action at+2